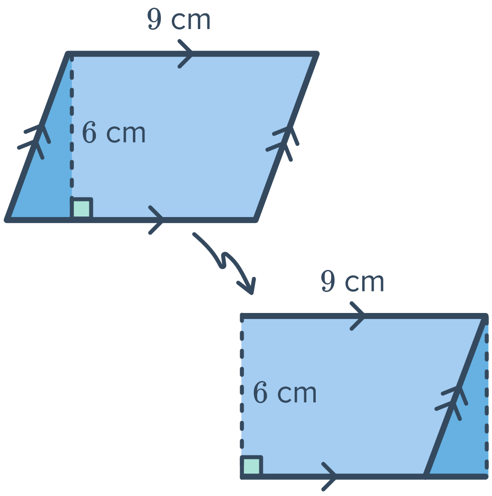
# Parallelograms

**1** The given parallelogram is formed into a rectangle:

**a** Find the area of the rectangle.

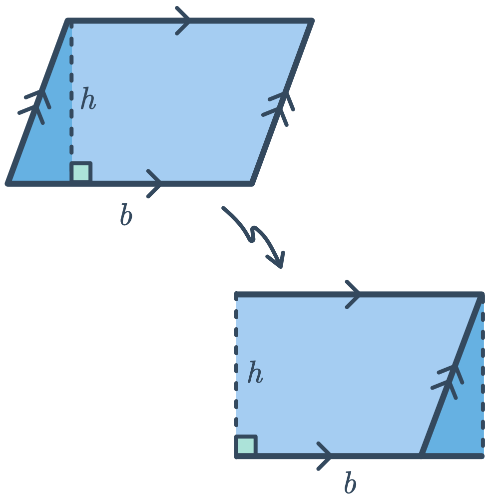
**b** Hence, find the area of the parallelogram.




**2** The given parallelogram is formed into a rectangle:

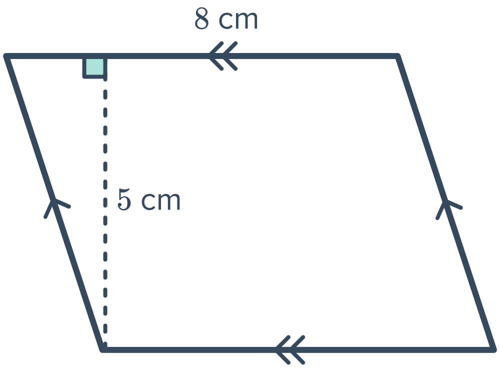
**a** Find an expression for the area of the rectangle in terms of $b$ and $h$.

**b** Hence, find an expression for the area of the parallelogram in terms of $b$ and $h$.

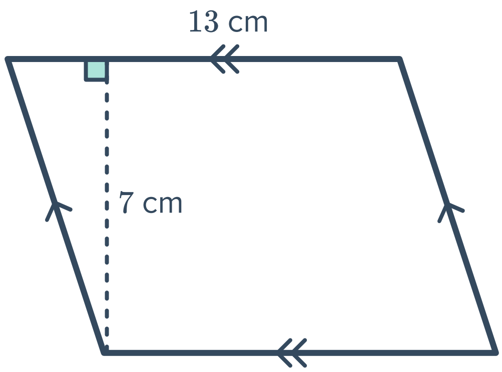



**3** Find the area of the following parallelograms:

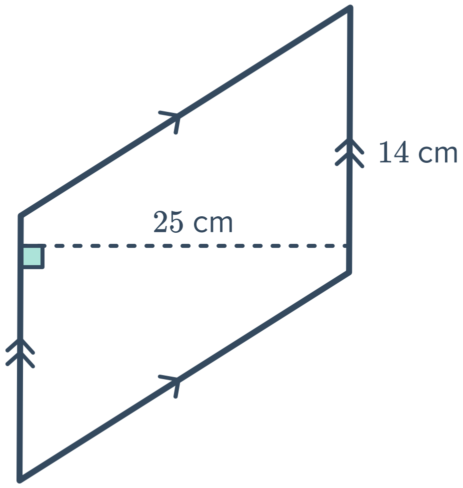
**a**



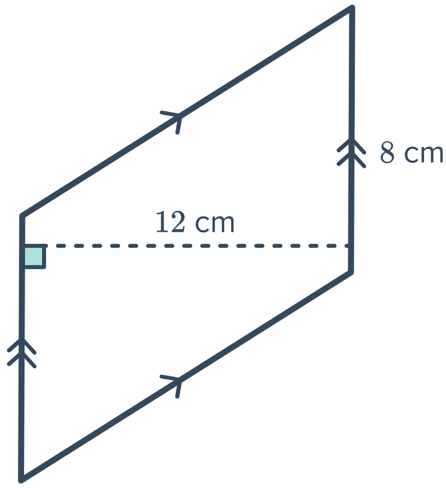

**b**



c




d




4 Determine whether the following pairs of values could be the dimensions of a parallelogram with an area of $70\,\text{mm}^2$.

a Base $= 10\,\text{mm}$, Height $= 7\,\text{mm}$

b Base $= 7\,\text{mm}$, Height $= 10\,\text{mm}$

c Base $= 1\,\text{mm}$, Height $= 70\,\text{mm}$

d Base $= 2\,\text{mm}$, Height $= 70\,\text{mm}$

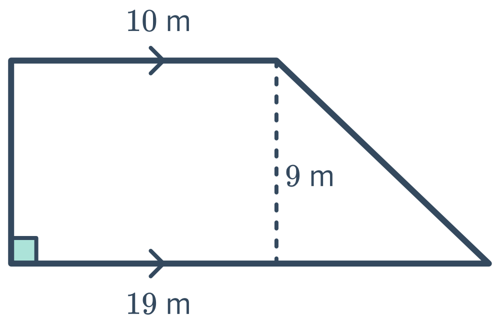
## Trapeziums

5 The given trapezium is split into a rectangle and a right-angled triangle:

a Find the area of the rectangle.

b Find the area of the triangle.

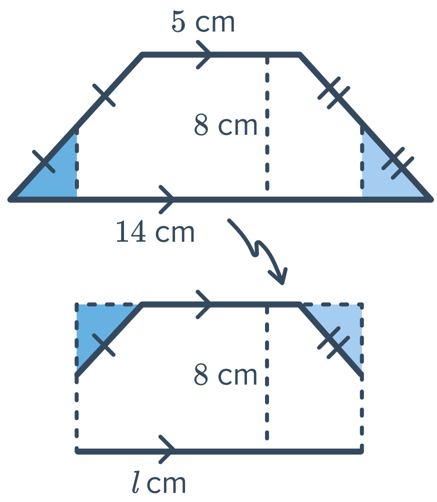
c Hence, find the area of the trapezium.



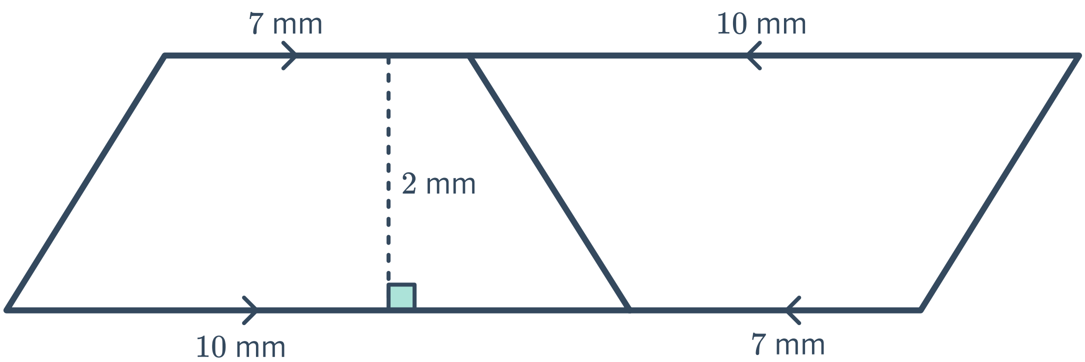
6 The given trapezium is formed into a rectangle:

a Find the length, $l$, of the rectangle.

b Hence, find the area of the trapezium.



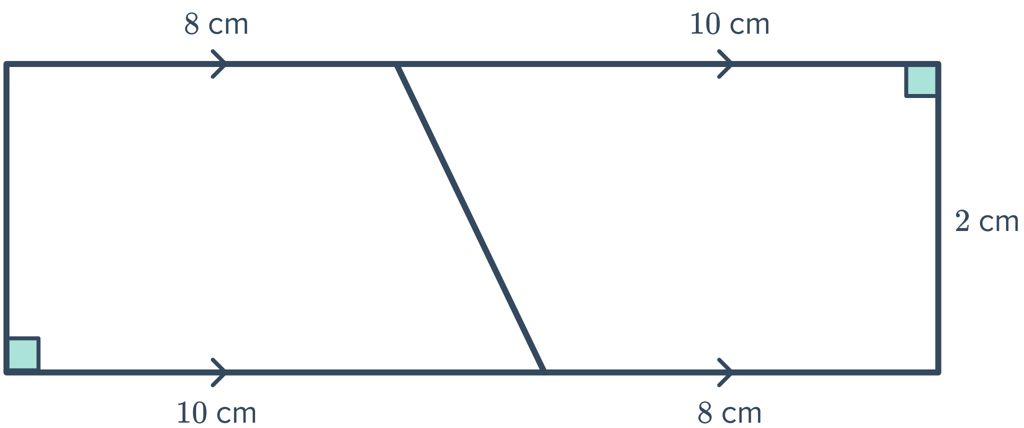

7 Two identical trapezia are put together to make a parallelogram:




a Find the area of the entire parallelogram.

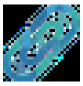
b Find the area of one of the trapezia.

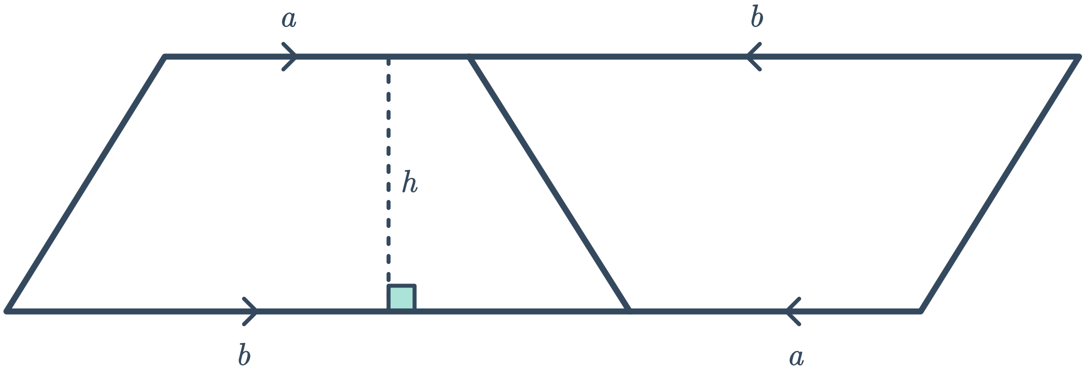
8 Two identical trapezia are put together to make a rectangle:




a Find the area of the entire rectangle.

b Find the area of one of the trapezia.

**9** Two identical trapezia are put together to  a parallelogram:

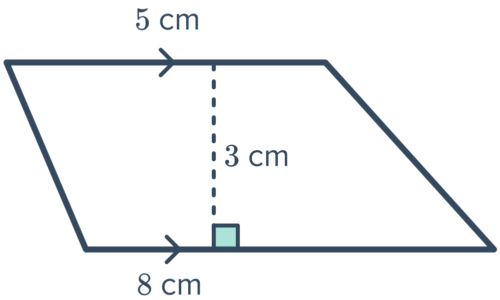



**a** Find an expression for the area of the entire parallelogram in terms of $a, b$ and $h$.

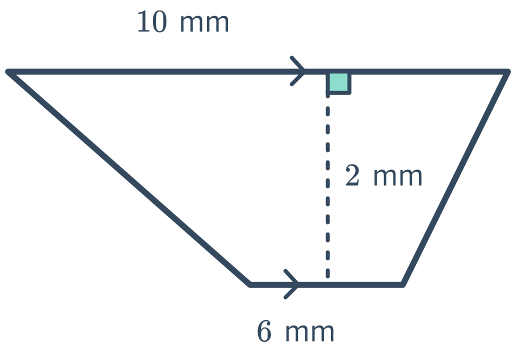
**b** Find an expression for the area of one trapezia in terms of $a, b$ and $h$.

**10** Find the area of the following trapeziums:

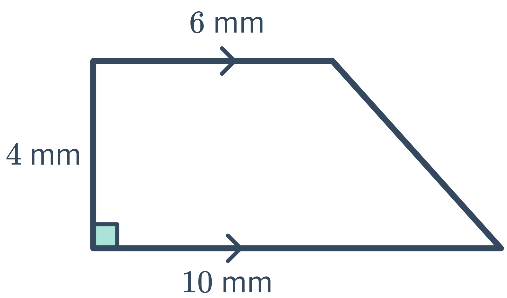
**a**



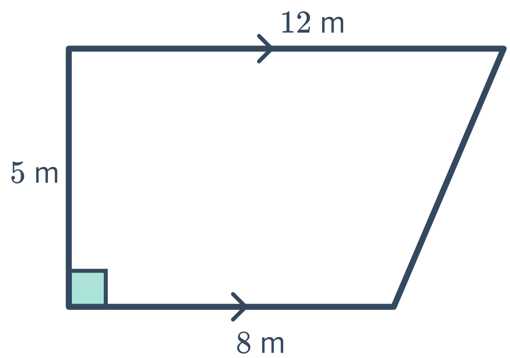

**b**



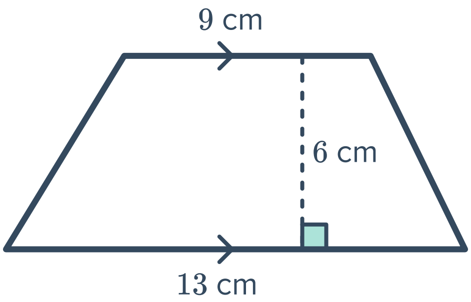

**c**



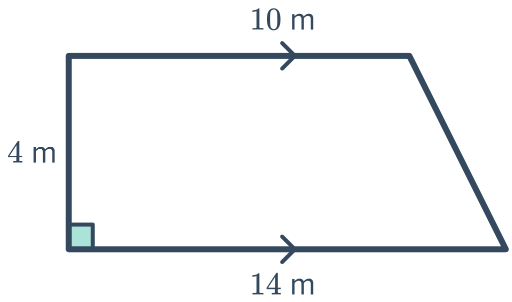

**d**



e



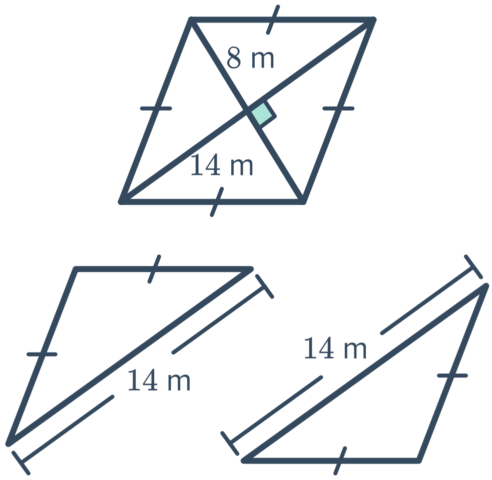

f




## Rhombuses

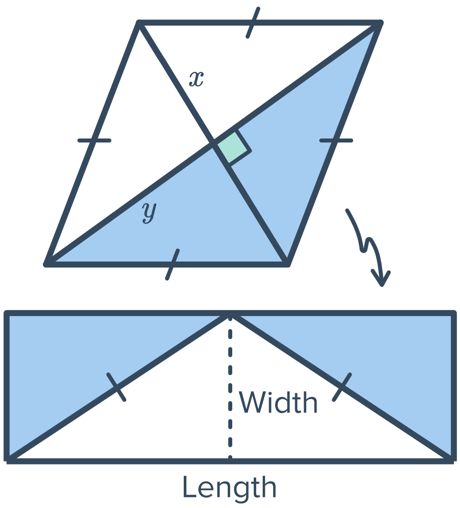
**11** The given rhombus can be split into two triangles:

- **a** Find the area of one triangle.
- **b** Hence, find the area of the rhombus.



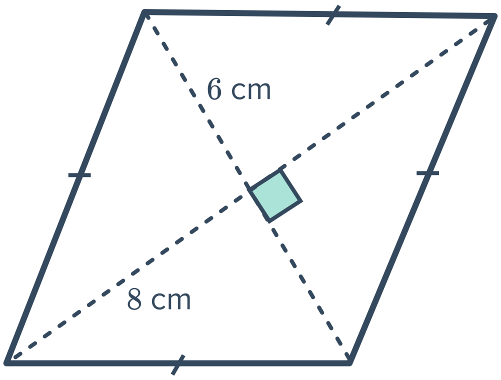

**12** The given rhombus is formed into a rectangle:

- **a** Find the length of the rectangle in terms of $y$.
- **b** Find the width of the rectangle in terms of $x$.
- **c** Find the area of the rectangle in terms of $x$ and $y$.
- **d** Hence, find the area of the rhombus in terms of $x$ and $y$.

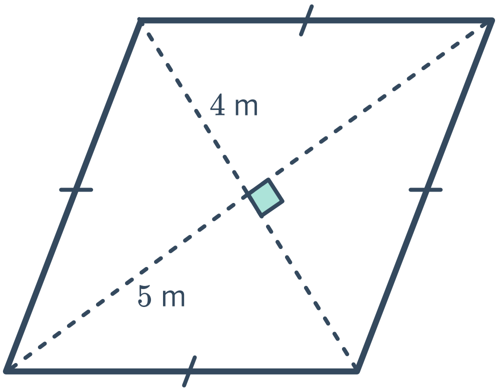


**13** Find the area of the following rhombuses:

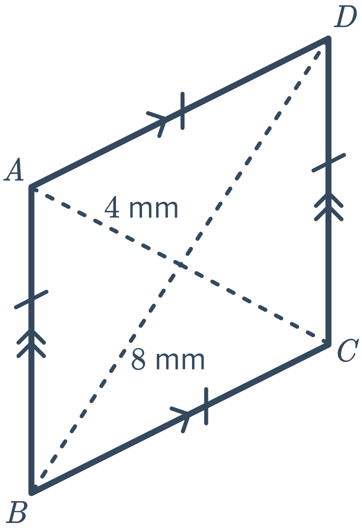
**a**



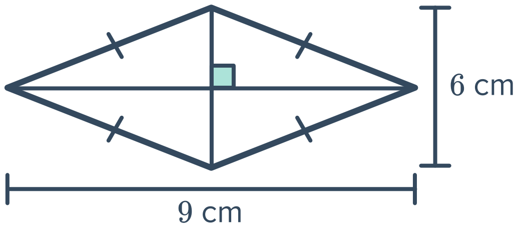

**b**



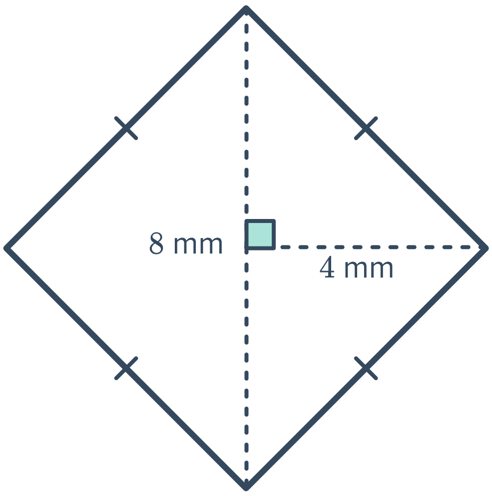

**c**



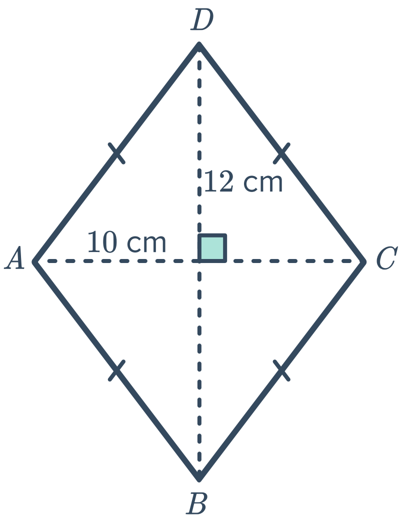

**d**




**e**




**f**



**14** Determine whether the following pairs of [redacted]s could be the diagonal lengths, $x$ and $y$ of a rhombus with an area of $9\,\mathrm{m}^2$.

**a** $x = 2\,\mathrm{m}$ and $y = 9\,\mathrm{m}$.

**b** $x = 6\,\mathrm{m}$ and $y = 3\,\mathrm{m}$.

**c** $x = 12\,\mathrm{m}$ and $y = 3\,\mathrm{m}$.

**d** $x = 6\,\mathrm{m}$ and $y = 6\,\mathrm{m}$.

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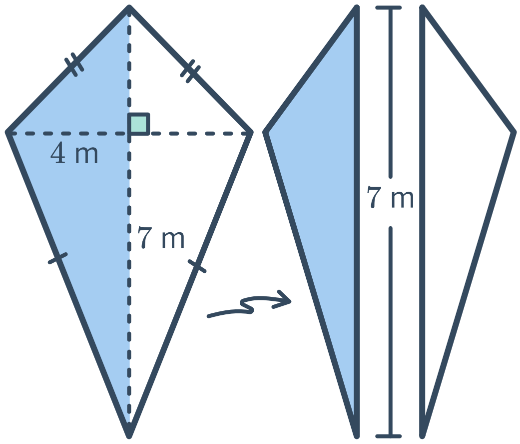
## Kites

**15** The given kite can be split into two triangles:

**a** Find the area of one of the triangles.

**b** Hence, find the area of the kite.

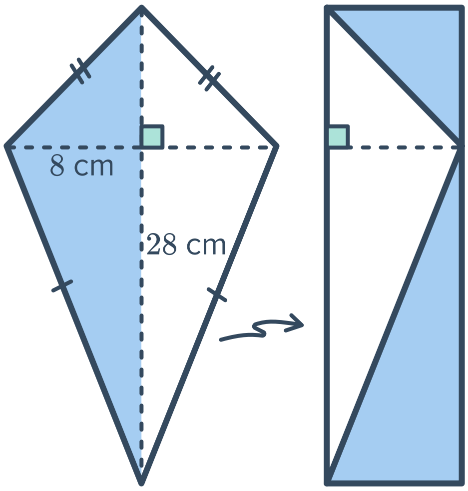



**16** The given kite is formed into a rectangle:

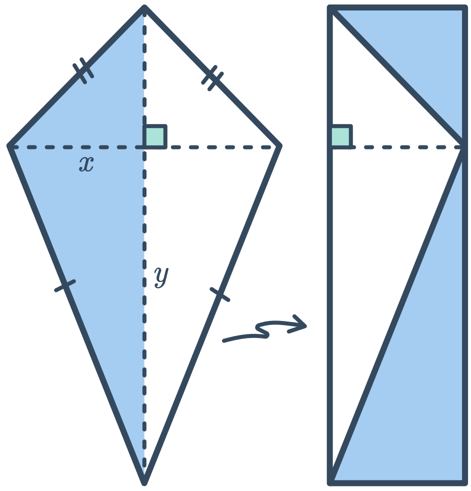
**a** Find the length of the rectangle.

**b** Find the width of the rectangle.

**c** Hence, find the area of the kite.



**17** The given kite is formed into a rectangle:




**a** Find the length of the rectangle in terms of $y$.

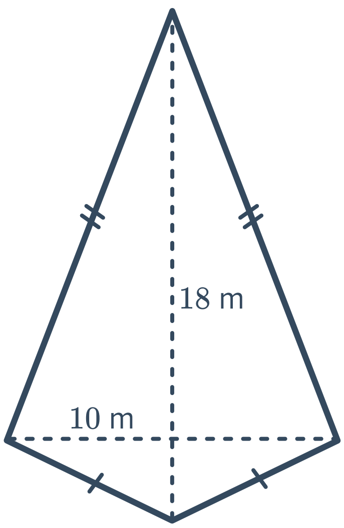
**b** Find the width of the rectangle in terms of $x$.

**c** Find the area of the rectangle in terms of $x$ and $y$.

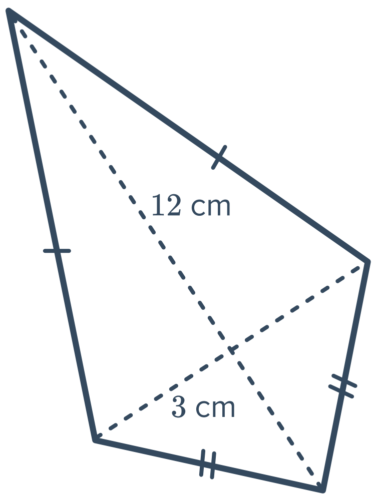
**d** Hence, find the area of the kite in terms of $x$ and $y$.

**18** Find the area of the following kites:

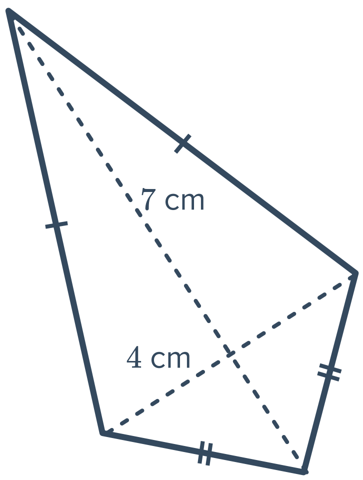
**a**



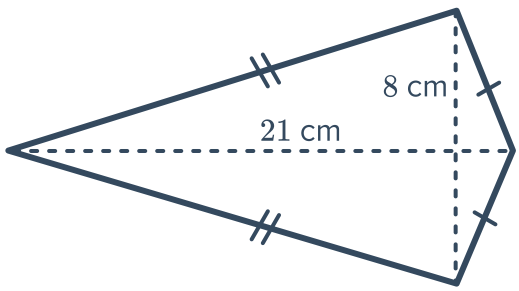

**b**



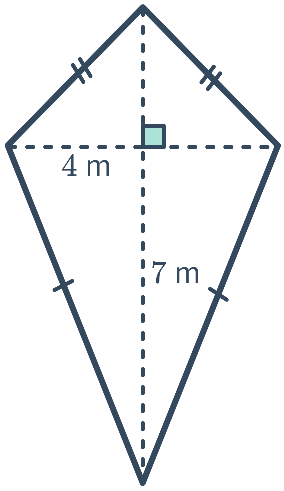

**c**



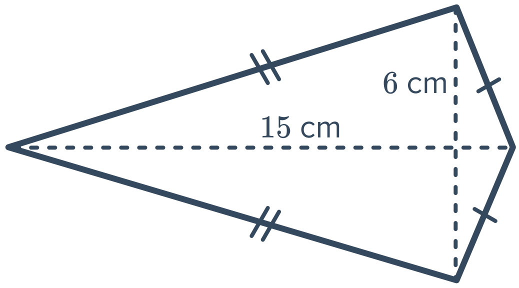

**d**



**e**



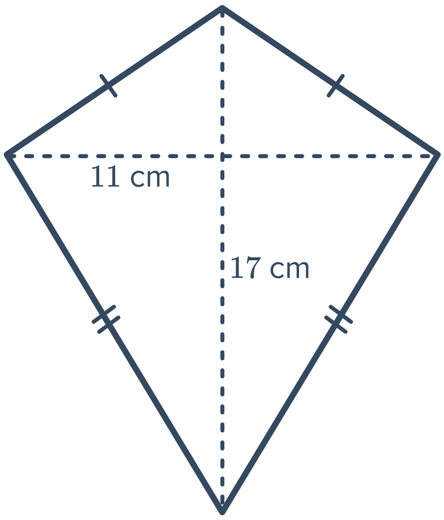

**f**



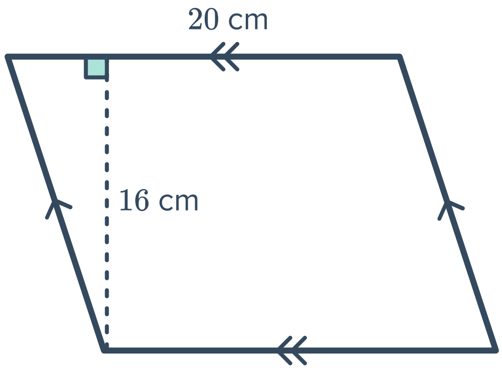

## Mixed areas

**19** Find the area of the following quadrilaterals:

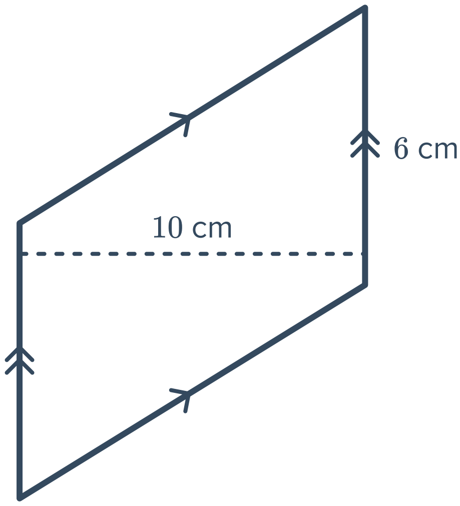
**a**



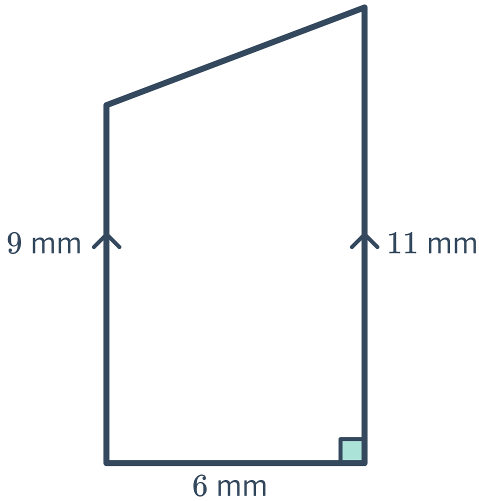

**b**



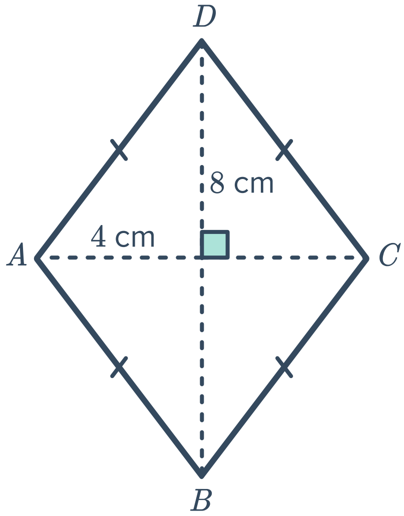

**c**



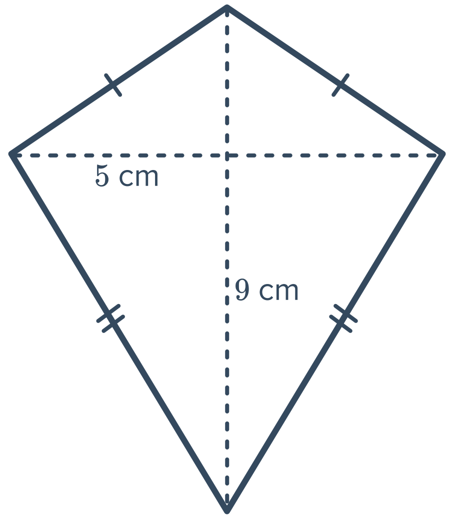

**d**



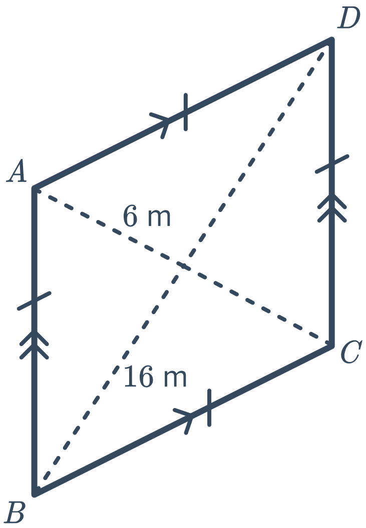
e


D
8 cm
4 cm
A
C
B

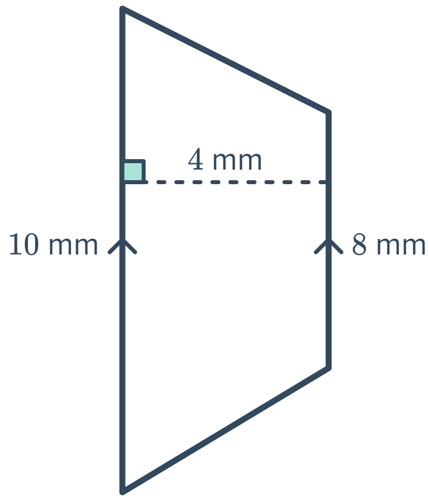
f


5 cm
9 cm

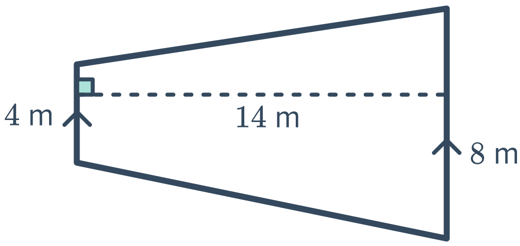
g


D
A
6 m
C
16 m
B

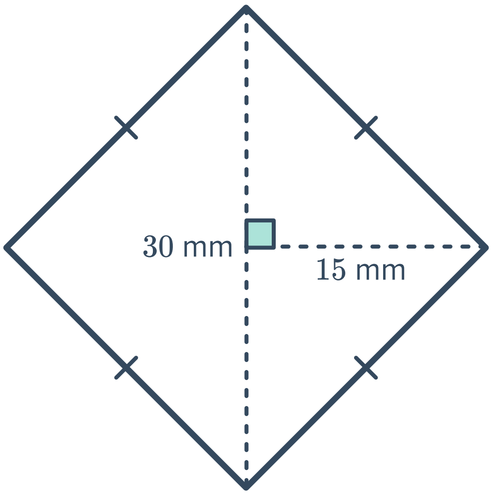
h


4 mm
10 mm
8 mm

i


4 m
14 m
8 m

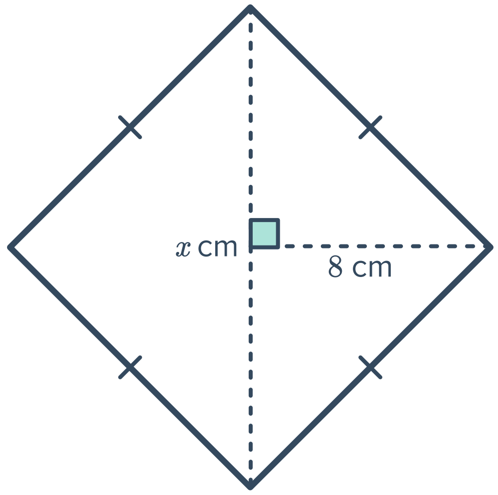
j


30 mm
15 mm

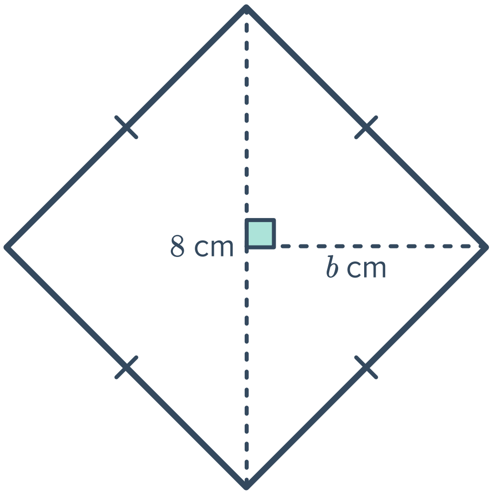
## Missing lengths

**20** For each of the following rhombuses, find the value of the pronumeral:

**a** $A = 64 \text{ cm}^2$

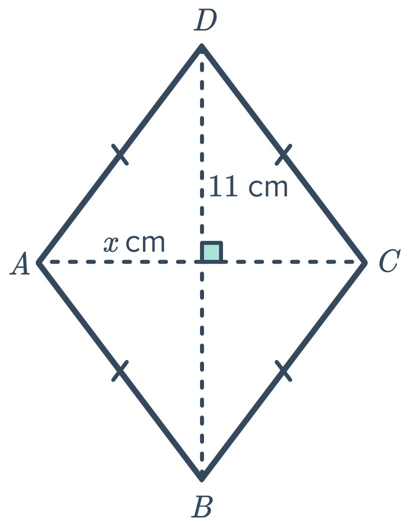


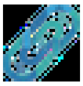

**b** $A = 128 \text{ cm}^2$



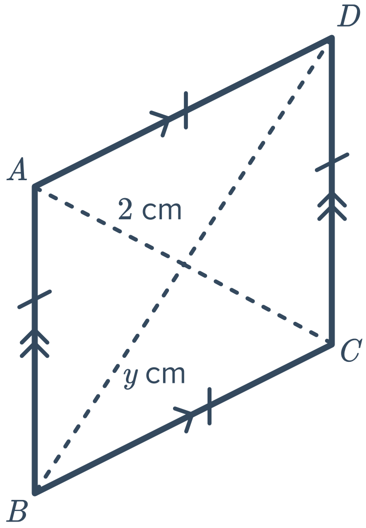

**21** Rhombus $ABCD$ has an area of $A = 55 \, \text{cm}^2$:

Given the diagonal $BD = 11 \, \text{cm}$, and $AC = x \, \text{cm}$, find the value of $x$.



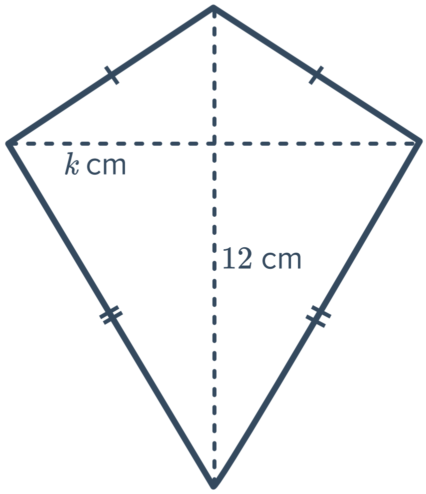
**22** Rhombus $ABCD$ has an area of $13\ \mathrm{cm}^2$: 

If diagonal $AC = 2$, and diagonal $BD = y$, find the value of $y$.



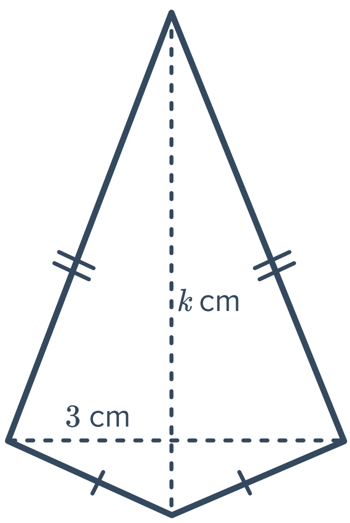

**23** The following kite has an area of $48\ \mathrm{cm}^2$.
The length of one of its diagonals is $12\ \mathrm{cm}$:

Find the length of the other diagonal, $k$.

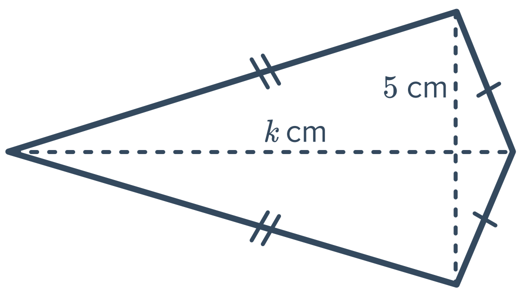



**24** For each of the following kites, find the value of $k$:

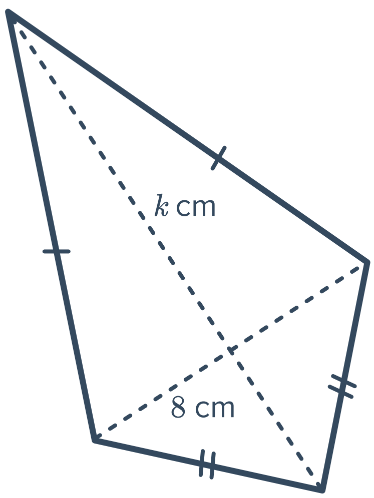
**a** $A = 15\ \mathrm{cm}^2$



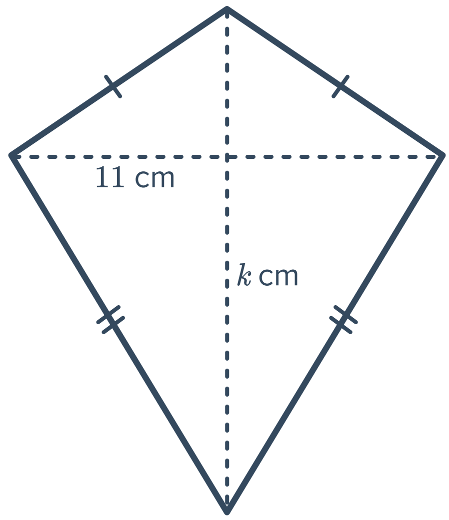

**b** $A = 22.5\ \mathrm{cm}^2$



c $A = 56\ \text{cm}^2$

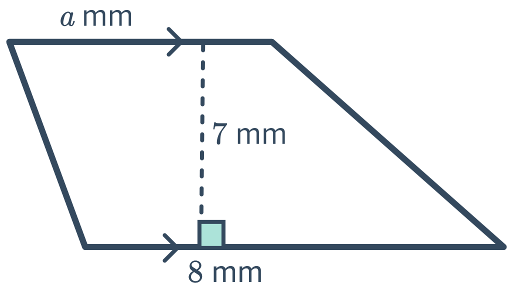



d $A = 137.5\ \text{cm}^2$

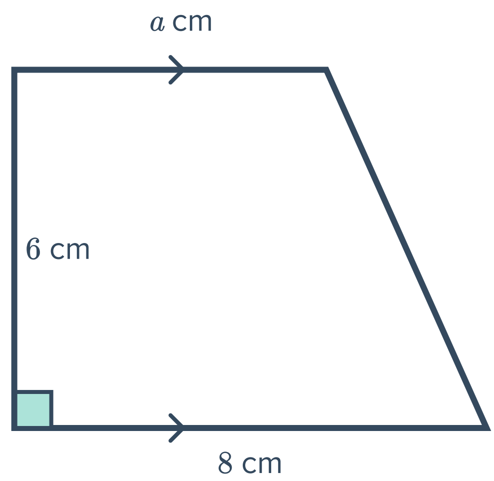



**25** For each of the following trapezia, find the value of the pronumeral:

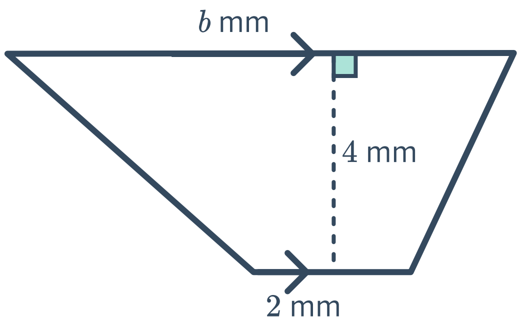
a $A = 42\ \text{mm}^2$



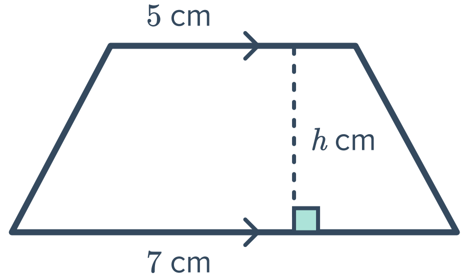

b $A = 36\ \text{cm}^2$



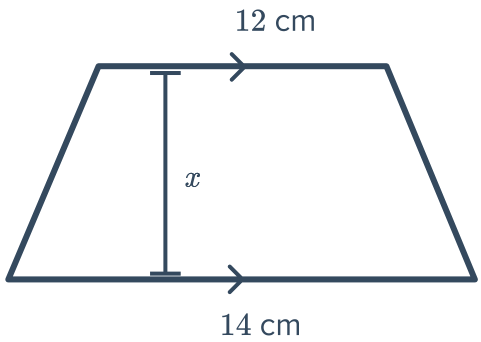

c $A = 20\ \text{m}^2$




d $A = 24\ \text{cm}^2$



**26** Find the value of $x$ if the area of the trapezium shown is $65\text{ cm}^2$:




**27** Find the perpendicular height, $h$, of a parallelogram that has an area of $45\,\text{cm}^2$ and a base of $5\,\text{cm}$.

**28** Find the base length, $b$, of a parallelogram that has an area of $216\,\text{mm}^2$ and a perpendicular height of $12\,\text{mm}$.

**29** The area of a kite is $640\text{ cm}^2$ and one of the diagonals is $59\text{ cm}$. If the length of the other diagonal is $y\text{ cm}$, find the value of $y$, rounded to two decimal places.

**30** Complete the table of base and height measurements for three parallelograms that all have an area of $24\,\text{m}^2$:

| Area $(\text{m}^2)$ | Base $(\text{m})$ | Height $(\text{m})$ |
|---|---|---|
| 24 | | 8 |
| 24 | 12 | |
| 24 | 6 | |

**31** Complete the table og the lengths of diagonal $x$ and diagonal $y$ for three kites that all have an area of $36\,\text{mm}^2$:

| Area $(\text{mm}^2)$ | Diagonal, $x$ (mm) | Diagonal, $y$ (mm) |
|---|---|---|
| 36 | | 18 |
| 36 | 24 | |
| 36 | 12 | |